

PATENT LANDSCAPE REPORT FOR HIGH-ACRYLONITRILE NBR-RELATED DISCLOSURES

1. ABSTRACT

This report synthesizes three primary patents retrieved under a high-acrylonitrile NBR search profile, with emphasis on how each disclosure relates to nitrile rubber composition control, degradation, or block-copolymer construction. The target material is high-acrylonitrile nitrile butadiene rubber (NBR), generally understood as a butadiene-acrylonitrile copolymer enriched in acrylonitrile to increase polarity, oil resistance, and gas-barrier performance relative to standard NBR. Among the selected patents, only US7875683B2 directly addresses nitrile rubber as the substrate and is therefore the most technically relevant to NBR processing; it discloses metathetic degradation of nitrile rubber using a metal-carbene metathesis catalyst combined with a transition-metal additive to reduce molecular weight before hydrogenation. US7592409B2 is relevant by composition and process analogy: it discloses controlled emulsion polymerization of block polymers containing an A block and a B block where the B block can include acrylonitrile and isoprene, and it provides detailed in-situ emulsification, RAFT/NMP-type control, and seed-latex strategies that are transferable to controlled nitrile-rubber architectures. US9296850B2 is only indirectly relevant, since it focuses on renewable isoprene production and polyisoprene/copolymer claims that merely list acrylonitrile among many possible comonomers; it does not disclose NBR synthesis. Across the landscape, the dominant process themes are controlled emulsion polymerization, in-situ surfactant generation, and metathesis-based molecular-weight reduction. The evidence suggests that high-ACN NBR development is most strongly supported by patents teaching precise monomer/comonomer control, aqueous emulsion processing, and post-polymerization molecular-weight tailoring rather than by direct high-ACN NBR examples.

2. METHODOLOGY

PRIMARY PATENT EVIDENCE

Patent 1

- Patent Number: US9296850B2
- Patent Title: Polymers of isoprene from renewable resources
- Assignee: Not disclosed

- Jurisdiction: US
- Publication Year: 2016
- Legal Status: Unknown
- Polymer Type: Not disclosed
- Relevance: Automatically extracted candidate

Polymerization / Synthesis Method - Synthesis method: This patent discloses microbial production of isoprene in engineered host cells, followed by recovery and polymerization of the isoprene into synthetic rubbers and other polymers. It also claims polyisoprene and isoprene copolymers, including copolymers where acrylonitrile is one of many optional comonomers, but it does not teach NBR-specific polymerization conditions. - Isoprene production threshold: more than about 0.002 percent of carbon available in the cell culture medium — Abstract - Polyisoprene delta-13C value: greater than -22% — Abstract - Polyisoprene delta-13C value: -30% to -28.5% — Detailed description - Polyisoprene delta-13C value: -34% to -24% — Detailed description - Liquid polyisoprene weight average molecular weight: 5,000 to 100,000 — Detailed description - Residual double bond content: 1 to 18% — Detailed description - Mooney viscosity: 55 to 120 ML 1+4 at 100°C — Background - Number average molecular weight: about 200,000 to 700,000 g/mol — Background - Lowest workable NBR feedstock Mooney viscosity: about 30 ML 1+4 at 100°C — Background - Hydrogenated nitrile rubber Mooney viscosity: in the order of 55 ML 1+4 at 100°C — Background - Hydrogenated nitrile rubber degree of hydrogenation: 50 to 100% — Background

Relevant Experimental Evidence - Example 1: Kudzu isoprene synthase was cloned and expressed in E. coli using pTrcHis2B, pET16b, and pCL1920-based constructs. - Example 4: Kudzu isoprene synthase was expressed in Bacillus subtilis using an aprE promoter-driven replicating plasmid. - Example 6: Yarrowia lipolytica expression vectors were constructed for kudzu and poplar isoprene synthase genes. - Example 11: Streptomyces albus expressing the codon-optimized isoprene synthase produced isoprene above control levels. - Example 12: E. coli strains using fatty acid or palm oil carbon sources produced mevalonate and isoprene.

Technical Relevance This patent is only indirectly relevant because it contains broad copolymer claims that list acrylonitrile as a possible comonomer, but the actual disclosure is centered on renewable isoprene biosynthesis. It does not provide a practical route to high-acrylonitrile NBR synthesis or composition control.

Patent 2

- Patent Number: US7875683B2
- Patent Title: Process for the metathetic degradation of nitrile rubber
- Assignee: Not disclosed
- Jurisdiction: US
- Publication Year: 2011
- Legal Status: Unknown
- Polymer Type: Not disclosed

- Relevance: Automatically extracted candidate

Polymerization / Synthesis Method - Synthesis method: This patent discloses reduction of nitrile rubber molecular weight by metathesis using a catalyst based on a transition-group 6 or 8 metal-carbene complex together with a transition-metal additive of formula $M(OZ)_m$. The process is intended to generate lower-molecular-weight nitrile rubber feedstocks that can subsequently be hydrogenated to HNBR with improved processability. - Transition metal additive formula: $M(OZ)_m$ — Claim 1 - Transition group for M in additive: 4, 5 or 6 — Claim 1 - m value: 4, 5 or 6 — Claim 1 - Z radical carbon atoms: 1-32 carbon atoms — Claim 1 - Z heteroatoms: 1 to 15 heteroatoms — Claim 1 - HNBR degree of hydrogenation: 50 to 100% — Background - HNBR Mooney viscosity: 55 to 120 ML 1+4 at 100°C — Background - HNBR number average molecular weight: about 200,000 to 700,000 g/mol — Background - Residual double bond content: 1 to 18% — Background - Lowest workable NBR feedstock Mooney viscosity: about 30 ML 1+4 at 100°C — Background - Hydrogenated nitrile rubber Mooney viscosity from such feedstock: in the order of 55 ML 1+4 at 100°C — Background - Target reduced Mooney viscosity before hydrogenation: less than 30 ML 1+4 at 100°C — Background - Target reduced number average molecular weight before hydrogenation: $M_n < 70,000$ g/mol — Background - Reported catalyst amounts in prior art: 307 ppm Ru and 61 ppm Ru based on nitrile rubber used — Background - Reported catalyst amount in prior art: 0.5 phr Grubbs (I) catalyst — Background - Reported catalyst amount in prior art: 614 ppm ruthenium based on nitrile rubber used — Background

Relevant Experimental Evidence - No experimental evidence extracted from available evidence.

Technical Relevance This is the most directly relevant patent in the set because it explicitly uses nitrile rubber (NBR) as the substrate and addresses its molecular-weight reduction. While it does not synthesize high-ACN NBR, it is highly relevant to downstream processing of high-acrylonitrile grades because composition-rich NBR often requires careful molecular-weight management to maintain processability.

Patent 3

- Patent Number: US7592409B2
- Patent Title: Styrene acrylonitrile isoprene triblock copolymer
- Assignee: Not disclosed
- Jurisdiction: US
- Publication Year: 2009
- Legal Status: Unknown
- Polymer Type: Not disclosed
- Relevance: Automatically extracted candidate

Polymerization / Synthesis Method - Synthesis method: This patent discloses controlled emulsion polymerization using in-situ generated emulsifiers, controlled free-radical agents, and seed-latex or microemulsion strategies to make block polymers. The disclosed block architectures include A-B-A triblocks where the B block can contain acrylonitrile and isoprene, making the

process conceptually relevant to controlled nitrile-containing elastomer synthesis. - Latent surfactant amount: about 0.2 grams — Example 1 - Styrene amount: 5 milliliters — Example 1 - Surfactant activator solution volume: 5 milliliters — Example 1 - Styrene charge: 1,000 grams — Example 2 - Oleic acid charge: 60.0 grams — Example 2 - Dibenzyltrithiocarbonate charge: 7.2 grams — Example 2 - RO water charge: 4,000 grams — Example 2 - Potassium persulfate charge: 40.0 grams — Example 2 - Tripotassium phosphate charge: 40.0 grams — Example 2 - Potassium hydroxide charge: 16.4 grams — Example 2 - Reaction temperature: 65°C — Example 2 - Complete conversion time: less than 1.5 hours — Example 2 - Final solids: 20.6% — Example 2 - Mn: 54,000 — Example 2 - PDI: 1.17 — Example 2 - Styrene charge: 1,000 grams — Example 3 - Hexadecyl half ester of maleic acid charge: 60.0 grams — Example 3 - Dibenzyltrithiocarbonate charge: 7.2 grams — Example 3 - RO water charge: 4,000 grams — Example 3 - Potassium persulfate charge: 40.0 grams — Example 3 - Tripotassium phosphate charge: 40.0 grams — Example 3 - Potassium hydroxide charge: 16.4 grams — Example 3 - Reaction temperature: 65°C — Example 3 - Complete conversion time: less than 3 hours — Example 3 - Final solids: 23.7% — Example 3 - Mn: 48,000 — Example 3 - PDI: 1.17 — Example 3 - 0.6 phr K/Oleate soap solution added — Example 3 - RO water charge: 200 grams — Example 8 - Styrene charge: 50 grams (0.48 moles) — Example 8 - Oleic acid charge: 3.0 grams (0.0106 moles) — Example 8 - Phenyl-t-butyl nitron charge: 0.44 grams (0.002485 moles) — Example 8 - Potassium persulfate charge: 2.06 grams (0.00762 moles) — Example 8 - Tripotassium phosphate charge: 2.13 grams (0.01 moles) — Example 8 - Potassium hydroxide charge: 0.9 grams (0.014 moles) — Example 8 - Reaction temperature: 65°C — Example 8 - Complete conversion time: about 7 hours — Example 8 - Mn: 35,000 — Example 8 - Mw: 61,000 — Example 8 - PDI: 1.74 — Example 8 - Styrene charge: 200 grams (1.923 moles) — Example 10 - Oleic acid charge: 12.0 grams (0.0424 moles) — Example 10 - Phenyl-t-butyl nitron charge: 2.0 grams (0.0113 moles) — Example 10 - Distilled water charge: 500 grams — Example 10 - Potassium persulfate charge: 8.0 grams (0.296 moles) — Example 10 - Tripotassium phosphate charge: 8.0 grams (0.0377 moles) — Example 10 - Potassium hydroxide charge: 3.6 grams (0.056 moles) — Example 10 - Reaction temperature: 75°C — Example 10 - Complete conversion time: about 3.5 hours — Example 10 - Mn: 39,000 — Example 10 - Mw: 48,000 — Example 10 - PDI: 1.22 — Example 10 - Reaction temperature: 65°C — Example 11 - Complete conversion time: about 7 hours — Example 11 - Mn: 41,000 — Example 11 - Mw: 70,000 — Example 11 - PDI: 1.70 — Example 11 - n-Butyl acrylate charge: 22.0 grams (0.171 moles) — Example 12 - Potassium persulfate charge: 0.5 grams (0.00184 moles) — Example 12 - Reaction temperature: about 65°C — Example 12 - Weak exotherm: about 70°C — Example 12 - Complete conversion time: about 2 hours — Example 12 - Mn: 45,000 — Example 12 - Mw: 77,000 — Example 12 - PDI: 1.70 — Example 12 - Benzaldehyde charge: 0.31 grams (0.00292 moles) — Example 13 - N-isopropylhydroxylamine charge: 1.4 grams of a 15% aqueous solution (0.0028 moles) — Example 13 - Reaction temperature: 75°C — Example 13 - Complete conversion time: about 4 hours — Example 13 - Mn: 31,000 — Example 13 - Mw: 55,000 — Example 13 - PDI: 1.76 — Example 13 - Dodecylbenzenesulfonic acid charge: 3.0 grams (0.0092 moles) — Example 14 - Sodium nitrite charge: 0.171 grams (0.002485 moles) — Example 14 - Reaction temperature: 75°C — Example 14 - Complete conversion time: about 3 hours — Example 14 - Mn: 97,000 — Example 14 - Mw: 147,000 — Example 14 - PDI: 1.51 — Example 14 - 1,1-Diphenylethylene charge: 0.448 grams (0.002485 moles) — Example 15 - Reaction temperature: 75°C — Example 15 - Complete

conversion time: about 2 hours — Example 15 - Mn: 51,000 — Example 15 - Mw: 92,000 — Example 15 - PDI: 1.80 — Example 15 - Styrene charge: 2,240 grams — Example 16 - Oleic acid charge: 134.4 grams — Example 16 - Dibenzyltrithiocarbonate charge: 16.0 grams — Example 16 - RO water charge: 3,808 grams — Example 16 - Potassium persulfate charge: 13.44 grams — Example 16 - Tripotassium phosphate charge: 89.6 grams — Example 16 - Potassium hydroxide charge: 36.96 grams — Example 16 - Reaction temperature: 65°C — Example 16 - Complete conversion time: less than 3 hours — Example 16 - Final solids: 38.9% — Example 16 - Mn: 45,500 — Example 16 - PDI: 1.11 — Example 16 - Sodium nitrite charge: 0.69 grams (0.01 moles) — Example 17 - Oleic acid charge: 3.0 grams (0.0106 moles) — Example 17 - Styrene charge: 50 grams (0.48 moles) — Example 17 - Acetic acid charge: 1.0 grams (0.0167 moles) — Example 17 - Distilled water charge: 2 grams — Example 17 - Potassium persulfate charge: 2.0 grams (0.00739 moles) — Example 17 - Tripotassium phosphate charge: 2.0 grams (0.01 moles) — Example 17 - Potassium hydroxide charge: 2.35 grams (0.0367 moles) — Example 17 - Reaction temperature: 75°C — Example 17 - Complete conversion time: about 2.5 hours — Example 17 - Final solids: 26.1% — Example 17 - Mn: 96,300 — Example 17 - Mw: 135,000 — Example 17 - PDI: 1.4 — Example 17 - Styrene charge: 50 grams (0.48 moles) — Example 18 - Oleic acid charge: 3.0 grams (0.0106 moles) — Example 18 - Crude 1-chloro-1,1-diphenylethane charge: approximately 0.01 moles — Example 18 - Distilled water charge: 200 grams — Example 18 - Potassium persulfate charge: 2.0 grams (0.0074 moles) — Example 18 - Tripotassium phosphate charge: 2.0 grams (0.0094 moles) — Example 18 - Potassium hydroxide charge: 1.55 grams (0.0242 moles) — Example 18 - Reaction temperature: 75°C — Example 18 - Complete conversion time: about 4 hours — Example 18 - Final solids: 22.7% — Example 18 - Mn: 40,000 — Example 18 - PDI: 1.37 — Example 18 - Styrene charge: 1,300 grams (12.5 moles) — Example 19 - Oleic acid charge: 78.0 grams (0.2756 moles) — Example 19 - Crude 1-chloro-1,1-diphenylethane charge: 39.0 grams — Example 19 - Deionized RO water charge: 2,470 grams — Example 19 - Potassium persulfate charge: 50.7 grams (0.1875 moles) — Example 19 - Tripotassium phosphate charge: 50.7 grams (0.2388 moles) — Example 19 - Potassium hydroxide charge: 29.5 grams (0.602 moles) — Example 19 - Reaction temperature: 75°C — Example 19 - Complete conversion time: about 5.5 hours — Example 19 - Final solids: 38.6% — Example 19 - Mn: 32,000 — Example 19 - PDI: 1.85 — Example 19 - 1-hexene charge: 25 grams — Example 20 - n-Butyl acrylate charge: 25 grams — Example 20 - Dipyrindyl charge: 0.38 grams — Example 20 - Oleic acid charge: 4.0 grams — Example 20 - Ethyl 2-bromoisobutyrate charge: 0.47 grams — Example 20 - KOH charge: 1.7 grams of 85% KOH in 139 milliliters water — Example 20 - Copper sulfate pentahydrate solution: 60.9 grams of a 1% aqueous solution — Example 20 - Hydrazine hydrate: 2 drops — Example 20 - Reaction temperature: about 67°C — Example 20 - Reaction time: 1.5 hours — Example 20 - Final solids: 10.9% — Example 20 - Isoprene incorporation: 5.5 mole% 1-hexene incorporated into the copolymer — Example 20

Relevant Experimental Evidence - Example 1: Demonstrated in-situ emulsification using latent surfactant/surfactant activator pairs in styrene. - Example 3: Showed RAFT-controlled emulsion polymerization with in-situ generated surfactant and a nitrile-rubber block architecture containing acrylonitrile and butadiene. - Example 5: Prepared S-NBR-S triblock copolymer latexes from seed latex. - Example 12: Demonstrated seed-latex reinitiation to form a block copolymer with n-butyl

acrylate. - Example 20: Showed ATRP emulsion copolymerization under microemulsion conditions and isolated a liquid copolymer.

Technical Relevance This patent is relevant because it explicitly discloses a block polymer B segment containing acrylonitrile and a conjugated diolefin, and it provides a controlled emulsion framework that can be adapted to nitrile-rubber synthesis. Although the examples are largely styrene-based, the process concepts are useful for controlling composition, architecture, and particle formation in high-acrylonitrile NBR systems.

3. CROSS-PATENT COMPARISON & SYNTHESIS TRENDS

- US7875683B2 is the only patent that directly uses nitrile rubber as the substrate; it focuses on post-polymerization molecular-weight reduction rather than monomer feed control.
- US7592409B2 provides the most relevant synthesis methodology for composition/architecture control, using in-situ emulsification, RAFT/NMP/ATRP-type control, and seed-latex strategies; it explicitly mentions acrylonitrile-containing B blocks.
- US9296850B2 is only tangentially related: it centers on renewable isoprene production and polyisoprene claims, with acrylonitrile appearing only as one optional comonomer in a broad claim list.
- No patent in the set discloses a direct high-acrylonitrile NBR copolymerization recipe with explicit acrylonitrile wt% feed ratios or final ACN content.
- Temperature windows in the relevant process patents cluster around 65-75°C for controlled emulsion polymerization in US7592409B2, while US7875683B2 emphasizes metathesis degradation followed by hydrogenation rather than polymerization temperature.
- US7875683B2 reports prior-art catalyst loadings in ppm Ru and phr catalyst terms, highlighting the catalyst-intensity issue in nitrile-rubber degradation; US7592409B2 instead uses persulfate/phosphate/KOH aqueous systems and organic control agents such as dibenzyltrithiocarbonate or PBN.
- US7592409B2 repeatedly achieves complete conversion within about 1.5 to 7 hours depending on recipe, whereas US7875683B2 discusses long reaction times as a limitation of prior metathesis degradation processes.
- For high-ACN NBR development, the most transferable evidence is controlled aqueous emulsion processing with surfactant generation and block-architecture control from US7592409B2, combined with molecular-weight tailoring concepts from US7875683B2.

4. CONCLUSION

The patent set does not contain a direct high-acrylonitrile NBR synthesis example, but it does provide two useful technical directions: controlled emulsion/block-copolymer methodology and nitrile-rubber molecular-weight reduction. For synthesis of high-ACN NBR, the strongest evidence

supports aqueous controlled polymerization with tight monomer-feed control, in-situ emulsification, and low-polydispersity control agents, while US7875683B2 indicates that post-polymerization metathesis can be used to tune molecular weight and processability of nitrile rubber feedstocks. Based on the evidence, the most defensible recommended approach is controlled emulsion copolymerization of butadiene with acrylonitrile using a surfactant-controlled aqueous system at roughly 65-75°C, with careful control of initiator and chain-transfer/control-agent loading, followed if needed by molecular-weight adjustment to maintain processability.

5. REFERENCES

- US9296850B2 | Polymers of isoprene from renewable resources | Not disclosed | US | 2016 | <https://patents.google.com/patent/US9296850B2/en>
- US7875683B2 | Process for the metathetic degradation of nitrile rubber | Not disclosed | US | 2011 | <https://patents.google.com/patent/US7875683B2/en>
- US7592409B2 | Styrene acrylonitrile isoprene triblock copolymer | Not disclosed | US | 2009 | <https://patents.google.com/patent/US7592409B2/en>